03.12.25

Gradient Descent

Linear Regression (Recap)

- Goal:
  - Given a training set, learn θ
  - Such that $h_\theta(x)$ is as close as possible to y

Least Mean Squares (LMS)

- For a dataset with n samples, minimize

$$J(\theta) = \frac{1}{2}\sum_{k=1}^{n}\left(h_\theta(x^{(k)}) - y^{(k)}\right)^2$$

Cost Function/Loss Function

$$h_\theta(x) = \sum_{i=0}^{d}\theta_i x_i = \theta^T x$$

Linear Regression

- Goal: Choose θ to minimize $\mathrm{J}(\theta)$

$$\theta^* = argmin_\theta J(\theta)$$

- How?

- Gradient Descent

Gradient Descent

Cost Function

$$J(\theta) = \frac{1}{2}\sum_{i=1}^{n}(h_\theta(x^{(i)}) - y^{(i)})^2$$

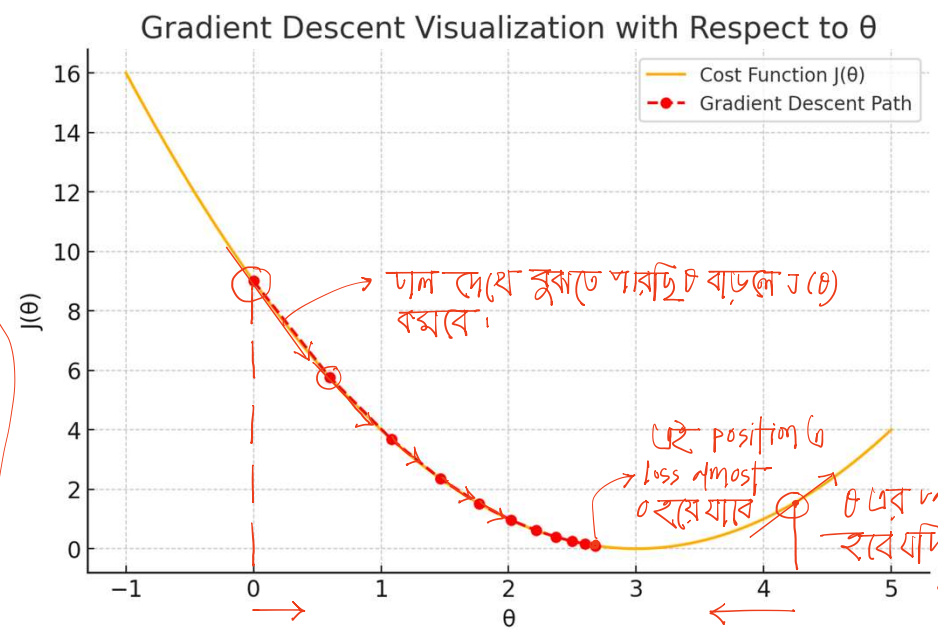
- Start from an initial θ
- Repeatedly update θ as follows

$$\theta_j := \theta_j - \boxed{\alpha \frac{\partial}{\partial \theta_j} J(\theta)}$$

θ এর প্রতিটা value র জন্য করতে হবে।
vectorized form এ লেখা যেতো।

- θ_j corresponds to the coefficient of the x_j
- α is called learning rate
  - Constant
  - Decaying

কেন?
→ সামনে discuss করা হবে



ঢাল দেখে বুঝতে পারছি θ বাড়লে J(θ) কমবে।

এই position এ loss almost 0 হয়ে যাবে

θ এর value কমাতে হবে যদি J(θ) কমাতে চাই

initially random θ দিয়ে start করবো।

iterative method.

gradient দেখে decision নেই θ বাড়াবো নাকি কমাবো, θ এর value কিভাবে কমাবো
- ঢাল (−ve) হলে θ বাড়াতে হবে
- ঢাল (+ve) হলে θ কমাতে হবে

actual minima তে gradient এর value 0 হবে, θ আর update হবে না। So এই process minima তে গিয়ে stop হবে।

Process টা থামাবো কখন (যেন overfit না হয়) → discussion সামনে আসবে।

Gradient Descent

$$\theta_j := \theta_j - \boxed{\alpha \frac{\partial}{\partial \theta_j} J(\theta)}$$

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{2} \sum_{k=1}^{n} \frac{\partial}{\partial \theta_j} \left(h_\theta(x^{(k)}) - y^{(k)} \right)^2$$

$$= \frac{1}{2} \sum_{k=1}^{n} 2 \left(h_\theta(x^{(k)}) - y^{(k)} \right) \frac{\partial}{\partial \theta_j} h_\theta(x^{(k)})$$

$$= \sum_{k=1}^{n} \left(h_\theta(x^{(k)}) - y^{(k)} \right) \frac{\partial}{\partial \theta_j} \left(\sum_{i=0}^{d} \theta_i x_i^{(k)} \right)$$

$$= \sum_{k=1}^{n} \left(h_\theta(x^{(k)}) - y^{(k)} \right) x_j^{(k)}$$

$J \to 0$ to d

$x_0 = 1$

Cost Function

$$J(\theta) = \frac{1}{2} \sum_{i=1}^{n} (h_\theta(x^{(i)}) - y^{(i)})^2$$

$\theta^T x$

$\sum_{i=0}^{d} \theta_i x_i$

Gradient Descent

$$J(\theta) = \frac{1}{2}\sum_{i=1}^{n}(h_\theta(x^{(i)}) - y^{(i)})^2$$

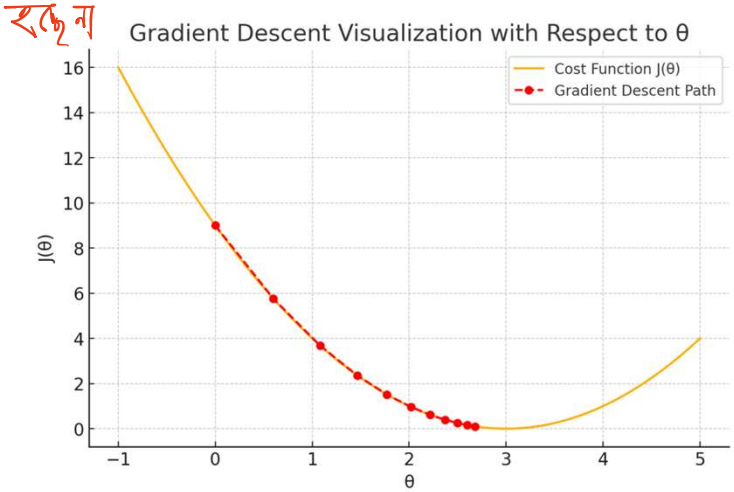
Cost Function

- Start from an initial θ

gradient zero/theta'র value update হচ্ছে না

Repeat until convergence

$$\theta_j := \theta_j + \alpha \sum_{k=1}^{n} \left(y^{(k)} - h_\theta(x^{(k)})\right) x_j^{(k)}, \quad \text{(for every } j\text{)}$$



কখন gradient-descent সবচেয়ে ভালো কাজ করবে?
→ convex function হলে।

maxima র প্রয়োজন হলে → cost function negative করে/gradient descent এর direction change করি।

gradient-ascent ও আছে। But করি না।

n এর value অনেক বেড়ে গেলে → summation still costly operation
d " _____ →

Reference